

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	_____
Project Title	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report presents FitBuddy, a web-based AI fitness planning application that generates personalized workout plans and nutrition or recovery tips based on user fitness goals. The system uses FastAPI with Google Gemini models to simplify structured fitness planning and allows users to refine plans through feedback.

Project Overview

Objective	The primary objective is to provide personalized and structured 7-day fitness plans based on user details such as age, weight, fitness goal, and preferred workout intensity.
Scope	The project covers AI-based workout generation, nutrition/recovery tips, feedback-based plan updates, user and plan storage, and an administrative view for monitoring users and plans.

Problem Statement

Description	Users may find it difficult to create structured workout routines that match their fitness goals and preferred intensity. Existing plans may also need manual changes when user preferences or progress change.
Impact	Providing personalized and adaptable plans can make fitness planning more structured, support consistent routines, and give users practical nutrition or recovery guidance.

Proposed Solution

Approach	FitBuddy uses Google Gemini models integrated with FastAPI to generate a personalized 7-day workout plan from user inputs. Gemini 1.5 Pro is used for workout generation and feedback-based plan updates, while Gemini Flash provides concise nutrition or recovery tips.
Key Features	• Personalized 7-day workout plan with warm-up, main workout, and cooldown/recovery guidance. • Nutrition or recovery tips based on the selected fitness goal. • Feedback-based workout plan updating. • SQLite database using SQLAlchemy for storing users and plans. • Admin view for registered users and original/updated plans. • FastAPI backend with HTML/Jinja2 frontend.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing resources	Standard system capable of running Python/FastAPI locally
Memory	RAM	Suitable for local development and application execution
Storage	Application, database and project files	Local disk storage

Software	Programming framework	Python, FastAPI, Uvicorn
Libraries	Required libraries	Jinja2, SQLAlchemy, SQLite, python-multipart, Google Generative AI SDK
Development Environment	Development tools	Python environment, pip/virtualenv or conda, Git
AI Service	Generative AI models	Google Gemini 1.5 Pro and Gemini Flash